

Algebra II with Trigonometry

Chapter 6.2

Mental Math (Gemini)

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[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Unit Circle Coordinates If the terminal point determined by a real number t on the unit circle is $(-\frac{3}{5}, \frac{4}{5})$, what is the exact value of $\sin t$?

Identifying Quadrants In which quadrant does the terminal point of t lie if $\tan t > 0$ and $\cos t < 0$?

Reciprocal Identities If $\sec t = 5$, what is the exact value of $\cos t$?

Even-Odd Properties (Sine) Given that $\sin t = 0.4$, what is the exact value of $\sin(-t)$?

Even-Odd Properties (Cosine) Given that $\cos t = -0.8$, what is the exact value of $\cos(-t)$?

Pythagorean Identity What is the exact numerical value of $\sin^2(42) + \cos^2(42)$?

Terminal Point Definitions In terms of the coordinates (x, y) on the unit circle, what is the mathematical ratio that defines $\cot t$?

Evaluating Functions at Multiples of Pi Based on the unit circle, what is the exact value of $\tan(15\pi)$?

Signs of Trigonometric Functions Which two trigonometric functions are positive when the terminal point lies in Quadrant IV?

Evaluating Displacement Formulas Given the oscillating mass displacement formula $y(t) = 4 \cos(3\pi t)$, what is the exact displacement when $t = 0$?